

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2019

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2019 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. A rectangular garden in Sakubva measures 14 m by 10 m. A path of width 1 m runs all around the outside. <<
>>(a) Area of the garden. [2]<<
>>(b) Area of the path. [3]<<
>>(c) A circular bed of radius 7 m is later dug. Area of the bed, $\pi = 22/7$. [3] **[8]**

2. (a) Make t the subject of $v = u + at$. [3]<<
>>(b) Simplify $(3x/4) + (x/6)$. [3]<<
>>(c) Solve $2/x = 5/(12)$. [4] **[10]**

3. 25 Form 4 pupils at Moleli scored:<<
>>6 (3 pupils), 6 (5 pupils), 2 (5 pupils), 4 (5 pupils), 3 (7 pupils).<<
>>(a) Modal mark. [1]<<
>>(b) Mean mark. [4]<<
>>(c) Range. [2]<<
>>(d) One extra pupil scores 6. Effect on the mean? [2] **[9]**

4. $AB \parallel CD$ with transversal EF in a street plan of Mutare.<<
>>(a) Angle $APQ = 52^\circ$. State the corresponding angle at Q , with a reason. [2]<<
>>(b) Find the co-interior angle to 52° . [2]<<
>>(c) A triangle has angles 52° , 49° and x° . Find x . [2]<<
>>(d) Classify the triangle (acute / right / obtuse). [2] **[8]**

5. Line $L: y = 3x + 5$.<<
>>(a) Gradient and y -intercept. [2]<<
>>(b) Coordinates where L meets the x -axis. [2]<<
>>(c) Show $A(2, 11)$ lies on L . [2]<<
>>(d) Equation of the line parallel to L through $(0, 4)$. [2] **[8]**

6. (a) Using ruler and compasses only, construct triangle ABC with $AB = 8$ cm, $BC = 7$ cm and $AC = 7$ cm. [4]<<
>>(b) Construct the perpendicular bisector of BC . [2]<<
>>(c) Mark the circumcentre and describe its property. [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** OA = a = (5 ; 3), OB = b = (-1 ; 3). M midpoint of AB. <<
>>(a) Vector AB. [3]<<
>>(b) |AB|. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show AM = MB. [3] **[12]**
- 8.** A kombi leaves Mutare at 08:00 and travels 120 km at constant speed, arriving after 3 hours. It stops 30 minutes, then returns at 80 km/h. <<
>>(a) Outward speed. [3]<<
>>(b) Time for the return. [3]<<
>>(c) Sketch a distance-time graph, labelling axes. [4]<<
>>(d) Average speed for the whole trip including the stop. [2] **[12]**
- 9.** (a) Solve $x^2 - 10x + 16 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Mutare is 8 m longer than it is wide. If the area is 20 m², form and solve an equation for the width w. [5]<<
>>(c) Sketch $y = (x - 2)(x - 8)$, showing intercepts. [3] **[12]**
- 10.** (a) y varies directly as x. When x = 4, y = 24. Find y when x = 10. [4]<<
>>(b) z varies inversely as x. When x = 3, z = 8. Find z when x = 6. [4]<<
>>(c) A farmer near Mutare has at most 40 hours. Maize takes 2 h/ha, vegetables 1 h/ha. Inequality for hectares m and v. [4] **[12]**
- 11.** Triangle P(2,1), Q(4,1), R(2,4). <<
>>(a) Reflect in the y-axis. Coordinates of P'. [4]<<
>>(b) Translate PQR by (3 ; 1). Give Q''. [3]<<
>>(c) Matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ maps (x;y). Describe fully and find the image of Q. [5] **[12]**
- 12.** A msasa tree is observed from A on level ground in Mutare. Angle of elevation of the top is 30°. Distance A to the foot is 30 m. <<
>>(a) Height of the object. Use $\tan 30^\circ = 1/\sqrt{3}$. [4]<<
>>(b) From B, 30 m due East of A, the bearing of the foot is 300°. Sketch. [3]<<
>>(c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**
- 13.** O is the centre. A, B, C on the circumference. AB is a diameter. Angle BAC = 40°. <<
>>(a) Angle ACB. Reason. [3]<<
>>(b) Angle BOC (centre, same arc BC). [3]<<
>>(c) D on the remaining circumference. Angle BDC. Reason. [3]<<
>>(d) State the angle-in-a-semicircle theorem. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).